

Prabhath Vinay Vipparthi

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SUMMARY

MS Data Science graduate from NJIT (GPA 3.7, May 2026) with production experience building NLP classification systems, machine learning pipelines, and large-scale data workflows. Built an explainable NLP classification system (FastAPI, spaCy) with full decision traceability, governance logging, and a 350+ test validation suite; benchmarked 6 ML models with SHAP explainability on a 20,000-record credit dataset; and engineered an LLM instruction-tuning pipeline (StarCoder2, vLLM) over 30,000+ source files on 5,791 quality-validated instruction-tuning pairs. Skilled in end-to-end ML development from feature engineering and model selection through production deployment and explainability reporting. work authorization: F-1 OPT, STEM-eligible.

EDUCATION

MS Data Science (Computational Track) — NJIT, Ying Wu College of Computing

May 2026 | Newark, NJ | GPA: 3.7

B. Tech Electrical & Electronics Engineering · Jawaharlal Nehru Technological University, Kakinada (JNTUK)

April 2023 | Andhra Pradesh, India

EXPERIENCE

Graduate Data Science Capstone — NJIT – Learning & Development Initiative (LDI)

Jan 2026 – May 2026 | Newark, NJ

- Engineered a production AI-assisted digital badge classification system (FastAPI, React/Vite, spaCy, SQLite) that automates NJIT's institutional taxonomy classification across 3 dimensions (category, type, cognitive level) achieving 100% classification accuracy on 20 real-world badge submissions validated against ground-truth labels from 5 institutional units (LDI, OSIL, Makerspace, NCE, OGI).
- Designed a 4-layer NLP extraction pipeline: 130+ lexicon phrase patterns, 44 regex rules, spaCy Bloom's taxonomy verb extraction, and an LLM stub handling OBv3 JSON, guided forms, and free-text inputs with robust signal coverage across all badge description styles.
- Built a deterministic 3-stage rule engine (19 rules: category → type → cognitive level) with full rule traceability, 8-element plain-English decision explanations per output, confidence scoring, immutable governance audit logs, and human-in-the-loop override workflows requiring written justification ensuring every classification is explainable and auditable.
- Delivered a 351-test automated validation suite (100% pass rate) across unit, integration, and 7 end-to-end workflow scenarios covering NLP extraction accuracy, full taxonomy classification consistency, and metadata ingestion correctness.

Office Assistant — New Jersey Institute of Technology

Sep 2025 – May 2026 | Newark, NJ

- Developed a web-based study room booking system to automate space reservation workflows for departmental staff and students, replacing manual scheduling processes.

PROJECTS

FinSight ? Pre-Market Intelligence Platform | FastAPI · PostgreSQL · pgvector · Gemini 2.5 Flash · GitHub Actions

- Built a live production pre-market briefing system ingesting ~130 equity symbols daily; computes RSI and MA signals, generates Gemini 2.5 Flash commentary, stores embeddings in pgvector, and emails subscribers via Resend. RAG layer answers plain-English queries over full briefing history with inline date citations.

NYC Taxi Medallion Data Pipeline | PySpark · dbt · DuckDB · Airflow · Streamlit

- End-to-end medallion pipeline over 5.97M NYC TLC Yellow Taxi trips (Jan/Feb 2024): raw Parquet ? PySpark cleaning ? dbt gold marts ? 23 data-quality tests ? daily Airflow DAG on DuckDB warehouse. Streamlit dashboard surfaces Manhattan at 75% of \$149.1M total revenue.

StarCoder2 Self-Alignment Pipeline ? LLM Instruction-Tuning | Python · vLLM · StarCoder2-3B · Hugging Face · Tree-sitter

- Processed 30,000+ TypeScript files from The Stack v2 using Tree-sitter AST parsing and TypeScript compiler type-checking; filtered to 5,791 quality-validated seed functions (~19% yield).
- Orchestrated S?C?I?R chain via StarCoder2-3B with vLLM (batch 32, T4 GPU); generated 5,791 concept annotations and 448 complete instruction-response pairs with per-output quality scoring for downstream fine-tuning.

Loan Approval Risk Prediction ? Explainable Credit Risk | Scikit-learn · TensorFlow · SMOTE · SHAP · GridSearchCV

- Classification pipeline on 20,000 loan applications (36 features, 76.1% class imbalance); SMOTE on training folds only; GridSearchCV-tuned 6-model comparison (LR, DT, RF, SVM, KNN, ANN). SHAP identified CreditScore, AnnualIncome, and DebtToIncomeRatio as top predictors for compliance-ready attribution reporting.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, TypeScript, Java

ML & Statistical Modeling: Scikit-learn, PyTorch, TensorFlow/Keras, SHAP, SMOTE, GridSearchCV, Cross-Validation

NLP & Text Analytics: spaCy, Hugging Face Transformers, vLLM, Regex Pipelines, Phrase Matching, Bloom's Taxonomy

Data Engineering: Hadoop, MapReduce, HDFS, Pandas, NumPy, ETL Pipelines, Tree-sitter

Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn

Cloud, DevOps & Infra: AWS (EC2, S3), Google Cloud, Docker, GitHub Actions, FastAPI, REST APIs

Databases: PostgreSQL, MySQL, SQLite

Tools & Environments: Git, Linux, Jupyter, VS Code, React, Colab